Soft Biometrics for Writer Identification: A Utility Law from Bangla Handedness

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***Abstract*—Writer identification from handwriting is a core task of forensic and historical document analysis, and soft biometrics such as handedness are often proposed to narrow its search; whether a *predicted* attribute actually helps, however, has rarely been tested. On the BanglaHand dataset (42 writers), we identified writers from Bangla paragraphs and strips with six representations in 120 settings and fused handedness evidence from a model that never saw the queried writer. Known handedness raised top-1 accuracy by 6.0 points, because identification errors were unrelated to hand. Predicted handedness (65.5% accurate) did not help: Bayesian fusion changed top-1 by -0.6 points and gallery filtering by -14.4. A utility law, top-1 with filtering = attribute accuracy $\times$ top-1 with known attribute, matched all 456 cases ($r = 0.996$) and implies that filtering needs about 87% accuracy, and a random-label control showed that attribute accuracy obtained on enrolled writers mainly measures writer recognition. The study gives the community a simple test of when a soft biometric is worth fusing with identification and a cheap check for leakage in reported attribute accuracies, the first such evidence for Bangla handwriting.**

***Index Terms*—Writer identification, Soft biometrics, Handedness, Bangla handwriting, Score fusion, Data leakage**

I. Introduction

Soft biometric traits such as gender, age or handedness do not identify a person, but they can complement a primary biometric by excluding candidates or reweighting scores [1], [2]. In handwriting, gender and handedness have been predicted from offline samples [3]–[9], often with the aim of narrowing the search of writer identification [10]. Whether this works depends on two numbers that are rarely reported together: how accurately the attribute is predicted for writers the model has never seen, and how often identification errors cross the attribute boundary. In forensic examination of Bangla documents, for example, a suspect's handedness may be known from the investigation, whereas that of the writer of a questioned document must be inferred from the handwriting itself.

A second problem is leakage. Handwriting identifies its writer very well [11], [12], so an attribute model trained on samples of the same writers it is applied to can recognise the writer and inherit the label [13]. In closed-set identification, where all gallery writers are known, this design is tempting.

We study both questions for handedness in Bangla, using a balanced, acquisition-controlled dataset [10] for which writer-independent handedness prediction has been characterised [14]. The contributions are: (i) a closed-set identification study with known and predicted handedness in 120 settings; (ii) a utility law that predicts the effect of attribute filtering from attribute accuracy, and the accuracy at which Bayesian fusion starts to help; (iii) a random-label control that separates attribute information from writer identity; and (iv) consequences for gallery prevalence and for the reporting of soft-biometric accuracies. Fig. 1 summarises the procedure.

II. Related work

Soft traits were proposed as auxiliary evidence for a primary biometric in personal recognition [1], and a survey covers their extraction from many modalities [2].

Gender has been predicted from on-line and off-line handwriting [3], [4], in an ICDAR 2013 competition with 194 participating teams [5], and together with other soft biometrics [6]. Handedness was predicted from Arabic and English texts of 121 writers with fuzzy feature reduction and a k-nearest-neighbour classifier [7], and with convolutional networks on the IAM and KHATT collections [8] and in later work [9]. Reported handedness accuracies range from F-measures of about 70% [7] to above 99% [9], under protocols that differ in whether test writers were seen in training [14]. These studies report attribute accuracy only.

Text-independent writer identification describes a writer by distributions of contour directions, hinges and graphemes [11] or by codebooks of contour fragments [12]. For Bangla, public collections support writer identification (BanglaWriting, 260 writers) [15] and character recognition (Boise State, 100 writers) [16], and writer verification reached 94.54% for 100 writers [17]. Table I summarises these works; none of the handwriting studies measured how a known or predicted attribute changes identification.

III. Data and identification

BanglaHand [10] contains 336 smartphone photographs of paragraphs by 42 writers (21 left-, 21 right-handed); each wrote six shared prompts (paragraphs 1–6) and two texts of their own choice (7–8) with shared materials, photographed in one session. Each paragraph is a *unit* ($n = 1$) and is also cut into $n = 2$, 4 or 8 horizontal strips of about 3.5, 1.75 and 0.9 lines (Fig. 1): the text block is divided into equal bands and each cut is moved to the emptiest row of the smoothed ink profile within half a line pitch. Strips keep the full width, and hence the scale, of the photograph. Of 5040 units, 39 strips with too little writing were removed.

Six representations from [14] are computed on every unit: writing-only STROKE features (208 contour-direction, hinge, curvature, stroke-ending and slant histograms), the HC-183

TABLE I

RELATED WORK. WRITERS: NUMBER OF WRITERS STATED IN THE CITED ABSTRACT OR RECORD; –: NOT STATED THERE. G: GENDER, H: HANDEDNESS, SB: SEVERAL SOFT BIOMETRICS.

Work	Task	Script / data	Writers	Method	Reported result
Jain et al. [1]	soft traits in personal recognition	primary biometric + soft traits	–	fusion with primary biometric	–
Liwicki et al. [3]	G	on-line and off-line handwriting	–	on-line and off-line features	–
Hassaïne et al. [5]	G (competition)	competition data (Kaggle)	–	194 participating teams	–
Siddiqi et al. [4]	G	off-line handwriting	–	handwriting features	–
Bouadjenek et al. [6]	SB	off-line handwriting	–	–	–
Al-Maadeed et al. [7]	H	Arabic, English	121	fuzzy feature reduction, k-NN	F-measure about 70%
Morera et al. [8]	G, H, G+H	English (IAM), Arabic (KHATT)	–	CNN	–
Rahmanian and Shayegan [9]	G, H	–	–	CNN	above 99%
Bulacu and Schomaker [11]	writer identification	off-line, text-independent	–	contour-direction, hinge, grapheme PDFs	–
Schomaker et al. [12]	writer identification	forensic and historic documents	–	contour-fragment codebooks	–
Mridha et al. [15]	dataset (writer ID, OCR)	Bangla (BanglaWriting)	260	–	–
Majid and Barney Smith [16]	dataset, character recognition	Bangla (Boise State)	100	bag of features, SVM	95.4% characters
Paul et al. [17]	writer verification	Bangla	100	handcrafted features, genetic selection	94.54% verification
[14]	H, writer-independent	Bangla (BanglaHand)	42	gradient descriptor, linear SVM	77.4% writer balanced acc.

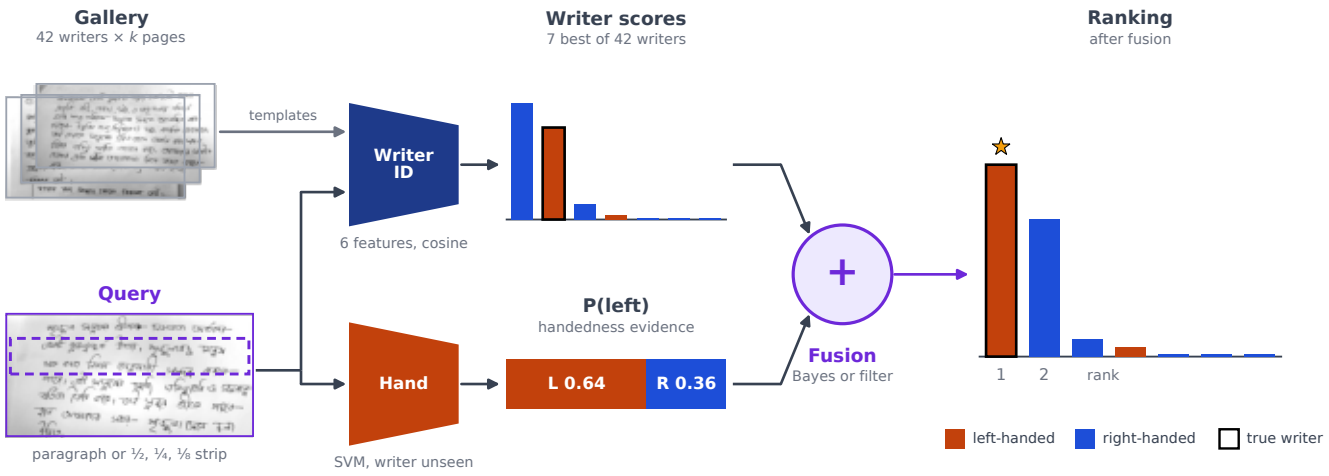


Fig. 1. Procedure for one real query (ViT-B/16, own-text paragraph, six references). The query, a paragraph or a strip (dashed: one of four), is scored against all 42 gallery writers (Writer ID; orange left-, blue right-handed, outlined the true writer) and given a probability of left-handedness by a model that never saw this writer (Hand); fusion combines both. Here the true writer (star) moved from rank 2 to rank 1; on average, fusion did not help (Section V-B).

gradient descriptor of the ink mask, frozen ImageNet ResNet-18 [18], EfficientNet-B0 [19] and ViT-B/16 [20] embeddings of grey tiles, and Stroke-CNN (ResNet-18 on binary ink tiles). For whole paragraphs they equal those of [14].

Identification is closed-set over all 42 writers. Features are standardised without labels and L2-normalised; a writer's template is the mean of k whole shared paragraphs, and the score s_w is the cosine similarity. Shared queries are matched against the other five prompts of every writer ($k = 5$), own-text queries against every subset of $k = 1$, 2, 3 or 6 prompts. Four query sizes, five reference conditions and six representations give 120 settings; the endpoint is top-1 accuracy, averaged over reference subsets.

IV. HANDEDNESS EVIDENCE AND FUSION

For every writer, a linear SVM ($C = 0.1$, as in [14]) on HC-183 is trained on shared units of the same size n from the other 41 writers and scores all units of the held-out writer; Platt scaling [21], fitted by inner five-fold cross-validation over the training writers, turns the scores into a log-odds ℓ of left-handedness. Sensitivity analyses use STROKE, HC-183 with ink density and photo intensity regressed out [14], or ℓ averaged over all own-text units of a writer.

Three scenarios are compared: *known*, with both hands known (for example, a suspect's hand known from the investigation); *query* (primary), with gallery hands known and the query hand predicted; and *both*, with gallery hands also predicted from the writer's references. With $P(h \mid x) = \sigma(\pm\ell)$, *Bayes* fusion ranks writers by $\log \mathrm{softmax}_w(s_w/T) + \log P(h_w \mid x)$,

Algorithm 1 Evaluation of one setting with predicted query handedness

Require: unit features, writer identities, hands h_v of the 42 writers, reference sets $\mathcal{R}$

for each writer w **do**
  train the SVM on shared units (size n) of all writers $v \neq w$
  fit Platt scaling by inner five-fold cross-validation over them
  $\ell_u \leftarrow$ calibrated left-handed log-odds of every unit u of w
for each reference set $R \in \mathcal{R}$ and query unit q **do**
  $t_v \leftarrow$ mean normalised features of R for every writer v
  $s_v \leftarrow \cos(x_q, t_v)$ for all writers v
for each writer w **do**
  fit T on the queries of writers $v \neq w$ (minimum NLL)
  rank writers by $\log \mathrm{softmax}_v(s_v/T) + \log P(h_v \mid \ell_q)$ for every query q of w
return top-1 per query; writer-level difference to no fusion

where h_w is the hand of writer w; *log-linear* fusion weights the second term by λ; and the *filter* ranks writers of the predicted hand first. T and λ are fitted on the queries of the other writers only. Algorithm 1 summarises one evaluation.

If the hand is predicted correctly with probability a, independently of identification, filtering keeps the true writer with probability a and then succeeds as often as with known hands:

$$\mathrm{top1}_{\mathrm{filter}} = a \cdot \mathrm{top1}_{\mathrm{known}}, \tag{1}$$

so filtering helps only if $a > \mathrm{top1}_{\mathrm{none}}/\mathrm{top1}_{\mathrm{known}}$. In words, a filter whose hand prediction is right 70% of the time discards the true writer in 30% of queries, and it pays off only if knowing the hand gains more than that loss. Bayesian fusion has no closed form, so we replaced ℓ by synthetic calibrated evidence of accuracy $a = 0.50, \ldots, 1.00$ (equal-variance Gaussian scores, log-odds $d\,z$ with $d = 2\Phi^{-1}(a)$; 200 draws). Prevalence was studied with galleries of the 21 right-handed and $m = 1$, 2 or 3 left-handed writers (1,000 draws), where the prior decision rule calls a query left-handed if $\ell + \log(m/21) > 0$.

For the leakage control, the SVM was also trained on the enrolled writers themselves (*enrolment-trained*): shared units were scored by five-fold cross-validation over paragraphs, so other paragraphs of the same writer were in training, and own-text units by the model fitted on all shared units. Both regimes were repeated with 20 random 21/21 writer splits as labels, each balanced for hand, so that accuracy and identification gain reached by a random label measure writer identity. The *both* scenario was controlled the same way and with random directions of HC-183 scaled to the spread of ℓ.

Differences in top-1 are computed per writer and tested with 9,999 sign flips over writers, with 2,000 writer-bootstrap intervals stratified by hand. Four primary tests, specified in advance, compare Bayes fusion (query scenario) with no fusion for ResNet-18 and ViT-B/16 on own-text paragraphs with one reference and on quarter strips with six references (Holm correction); all other p-values, and counts of significant settings, which share writers, are descriptive. A feasibility check on saved paragraph features, which covered the paragraph-level primary settings, preceded the protocol.

TABLE II
WRITER-INDEPENDENT HANDEDNESS ACCURACY (%, SHARED / OWN-TEXT UNITS) BY QUERY SIZE, AND WRITER-LEVEL ACCURACY (PARAGRAPH UNITS AVERAGED PER WRITER). LINEAR SVM, LEAVE ONE WRITER OUT.

Model	Paragraph	$\frac{1}{2}$ strip	$\frac{1}{4}$ strip	$\frac{1}{8}$ strip	Writer
HC-183	68.7 / 65.5	65.9 / 66.1	65.9 / 66.9	63.2 / 61.7	73.8
HC-183 adj.	56.7 / 61.9	63.3 / 57.1	58.6 / 60.9	60.8 / 59.3	61.9
STROKE	53.2 / 48.8	51.6 / 46.4	44.6 / 41.2	46.6 / 46.7	54.8

V. RESULTS

A. Known handedness complements identification

Writer-independent handedness was predicted with 68.7% unit accuracy on shared paragraphs, 65.5% on own texts and 73.8% at the writer level, and with 61.7–66.9% on strips; adjusted HC-183 (61.9%) and STROKE (48.8%) were weaker (Table II). No model or query size came close to the break-even accuracies of Section V-C. Among identification features, STROKE was strongest (68.7% top-1 on one-line own-text strips with one reference), while Stroke-CNN collapsed on strips (10.7% at $n = 4$).

Identification errors were unrelated to hand: 44–53% of misidentified queries went to a writer of the other hand, against 51.2% expected by chance (Fig. 2c). Known handedness therefore removed about half of the competitors and raised top-1 by 6.0 points on average (up to 14.9), significantly in 106 of 120 settings (Table III). Accuracy fell as the query shrank and rose with more reference pages, and the gain from a known hand was largest where identification was weakest, whereas fusion with the predicted hand stayed at or below the baseline throughout (Fig. 2a,b).

B. Predicted handedness does not help

None of the four primary tests showed a gain (differences -1.5 to -0.3 points, $p \geq 0.37$, Holm $p = 1$; Table IV). Over all settings, Bayes fusion changed top-1 by -0.6 points on average (significant gains in 0, losses in 1 settings), log-linear fusion by -0.2, as its weight was fitted near zero, and the filter by -14.4 points (losses in 89 settings).

C. A utility law

Eq. (1) predicted the filter accuracy of all 456 combinations of setting and handedness model ($r = 0.996$, mean absolute error 1.5 points, bias -0.9; Fig. 3a). The break-even accuracy for own-text queries was 87% (median; range 56–98%), far above the 65.5% achieved. With synthetic calibrated evidence (Fig. 3b), Bayes fusion gained at least half a point from an accuracy of 60–80%, at most 2.7 points at the achieved accuracy, and 0.7–9.8 points at 95%. The real evidence fell 1.3 points short of synthetic evidence of equal accuracy, so it is less well calibrated or not independent of identification. Applying the filter only to the most confident 30% of predictions (75% accurate, still below break-even) changed top-1 by -8.7 to +0.4 points.

When left-handed writers were rare (one to three in the gallery; Fig. 3c), Bayes fusion raised top-1 of left-handed

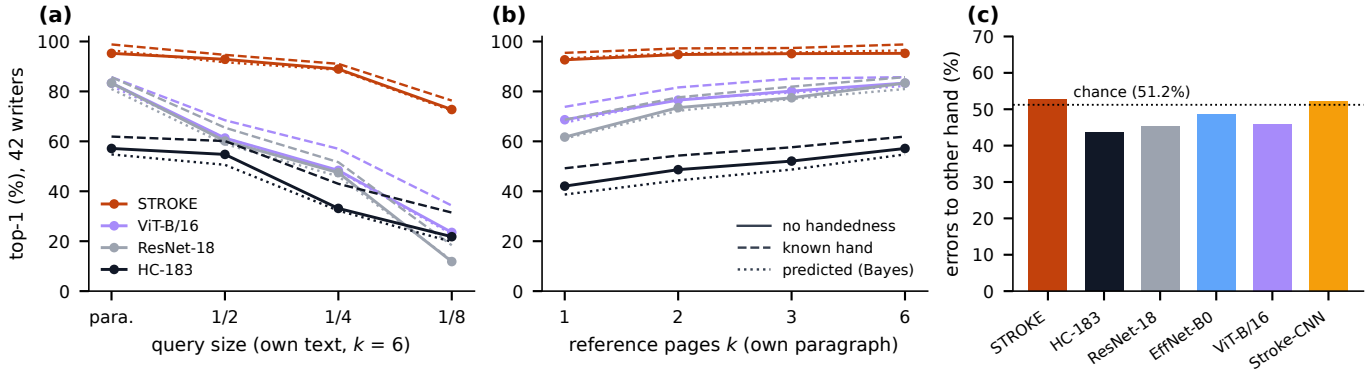


Fig. 2. Identification with and without handedness. Top-1 accuracy (a) as the own-text query shrinks from a paragraph to about one line (six references) and (b) as the number of reference pages grows (own-text paragraph queries): no handedness (solid), known hand (dashed), predicted hand with Bayes fusion (dotted). (c) Share of identification errors whose top-ranked writer has the other hand, over all settings of each representation; dotted: share expected if errors were unrelated to hand.

TABLE III

TOP-1 IDENTIFICATION ACCURACY (%) WITHOUT HANDEDNESS (NONE), WITH KNOWN HANDEDNESS (KNOWN) AND WITH PREDICTED QUERY HANDEDNESS (BAYES FUSION; FILTER): OWN-TEXT PARAGRAPH AGAINST ONE SHARED REFERENCE (MEAN OVER THE SIX CHOICES), OWN-TEXT QUARTER STRIP AGAINST SIX REFERENCES, AND SHARED QUARTER STRIP AGAINST THE OTHER FIVE PROMPTS. HANDEDNESS: HC-183, LEAVE ONE WRITER OUT.

	Own paragraph, $k = 1$				Own quarter strip, $k = 6$				Shared quarter strip, $k = 5$			
Representation	None	Known	Bayes	Filter	None	Known	Bayes	Filter	None	Known	Bayes	Filter
STROKE	92.7	95.4	93.3	62.1	89.0	91.0	88.7	60.0	96.0	96.9	95.7	63.7
HC-183	42.1	49.2	38.7	28.0	33.1	43.0	32.2	25.7	42.4	52.1	41.1	34.2
ResNet-18	61.7	68.7	61.1	42.9	47.5	51.6	46.0	35.8	63.4	69.7	62.7	45.7
EffNet-B0	55.6	62.5	53.8	38.7	49.3	57.6	49.0	38.2	63.7	70.5	63.5	45.0
ViT-B/16	68.7	73.8	67.5	45.2	48.4	57.0	48.1	39.1	64.8	72.2	64.7	45.8
Stroke-CNN	58.9	70.6	60.1	43.8	10.7	15.5	10.7	8.4	15.8	23.9	16.6	14.3

TABLE IV

PRIMARY TESTS: BAYES FUSION OF PREDICTED QUERY HANDEDNESS AGAINST NO FUSION (TOP-1, %). DIFF.: WRITER-LEVEL MEAN DIFFERENCE (POINTS) WITH WRITER-BOOTSTRAP INTERVAL; p: 9,999 SIGN FLIPS; HOLM OVER THE FOUR TESTS.

Setting	Representation	None	Bayes	Diff. [95% CI]	p	Holm
Own paragraph, $k = 1$	ResNet-18	61.7	61.1	−0.6 [−4.0, +2.2]	0.84	1.00
Own paragraph, $k = 1$	ViT-B/16	68.7	67.5	−1.2 [−3.6, +1.2]	0.48	1.00
Own strip ($n = 4$), $k = 6$	ResNet-18	47.5	46.0	−1.5 [−4.2, +0.9]	0.37	1.00
Own strip ($n = 4$), $k = 6$	ViT-B/16	48.4	48.1	−0.3 [−3.6, +2.7]	1.00	1.00

queries by 2.7 points on average (up to 14.3 for weak representations) and lowered that of right-handed queries by at most 1.3. A filter using the prior decision rule, however, lowered top-1 of left-handed queries (ViT-B/16) from 78.6% to 2.1–14.5%: an attribute model of this accuracy cannot overcome a low prior.

D. Leakage control

Trained on the enrolled writers, the handedness model reached 82.1% on own-text paragraphs, 16.7 points more than for unseen writers (65.5%; Fig. 4a). Random labels reached 52.4–76.2% (mean 62.6%, 12.6 points above chance) under the same training, but fell below chance for unseen writers (33.3%), the known bias of leave-one-out on uninformative labels. Most of the accuracy gained by enrolment training therefore reflects the writer, not the hand. In identification (Fig. 4b), enrolment-trained handedness changed top-1 by +1.1 points and random labels by +0.1.

E. Predicted hands on both sides

In an exploratory analysis with gallery hands also predicted, fusion raised top-1 by 0.95 points on average (significant in 29 settings, never lower). Random labels changed top-1 by +0.03 points, scaled random labels by +0.21 ($p = 0.048$) and random directions by +0.19 ($p = 0.010$; Fig. 4c). Agreement of predicted hands thus adds a little information that random projections do not, best read as agreement on a writer-consistent, handedness-related trait.

F. Comparison with previous work

Table V compares the scope of this study with earlier work. Earlier studies either predicted attributes [3]–[9] or identified writers without them [11], [12], [17]. To our knowledge, this is the first study to address all columns of the table, and the first for Bangla.

VI. DISCUSSION

Handedness is complementary to writer identification: features from contour hinges to ImageNet embeddings made

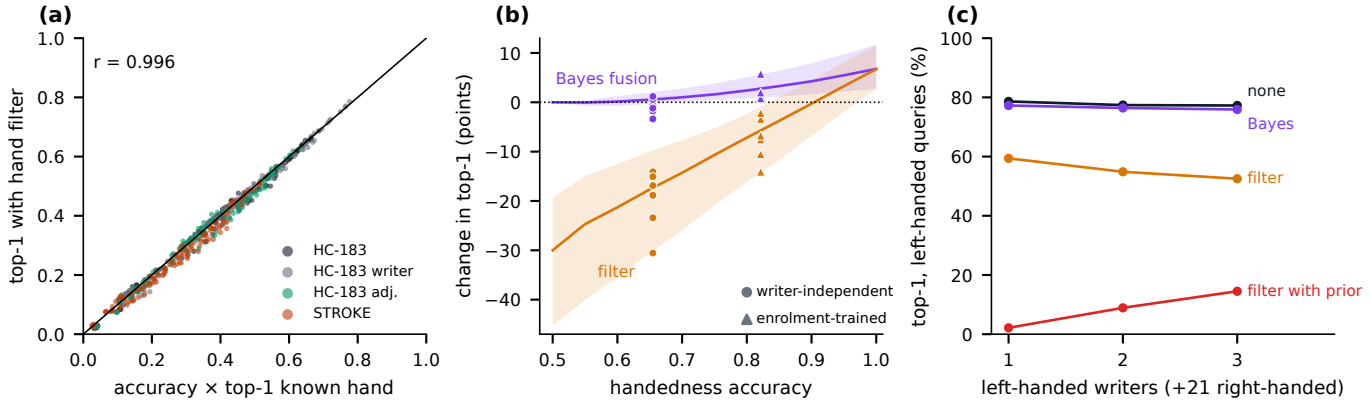


Fig. 3. (a) Utility law: observed top-1 with hand filtering against handedness accuracy × top-1 with known handedness (Eq. 1), for every setting and handedness model. (b) Change in top-1 against handedness accuracy for synthetic calibrated evidence (lines: mean over representations; band: range; own-text paragraphs, one reference) and for real evidence (circles: writer-independent; triangles: enrolment-trained). (c) Galleries of 21 right-handed and 1–3 left-handed writers: top-1 of left-handed queries (ViT-B/16, own-text paragraphs, one reference).

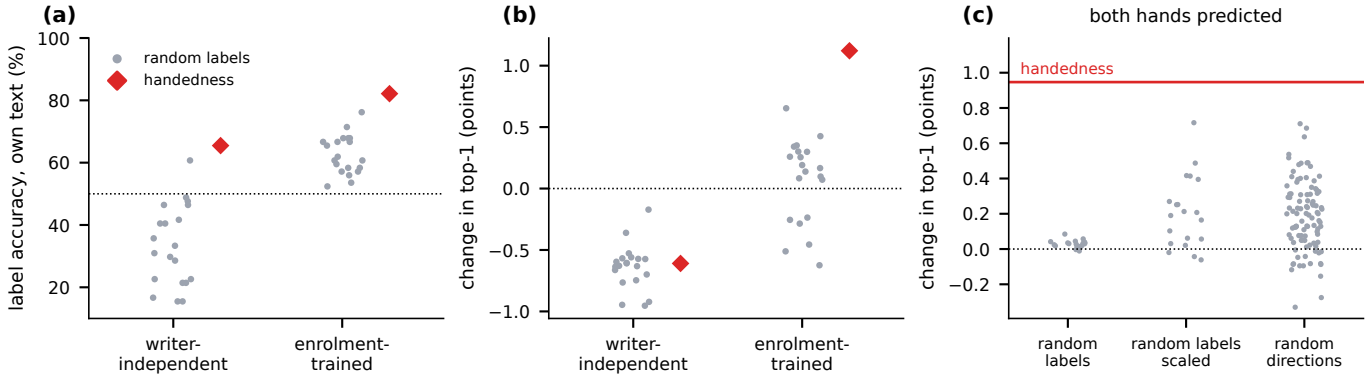


Fig. 4. Random-label control. (a) Accuracy on own-text paragraphs of handedness (red) and of 20 random hand-balanced labels (grey), for models that never saw the writer (writer-independent) or were trained on the enrolled writers. (b) Mean change in top-1 over all settings with this evidence fused (Bayes, gallery labels known). (c) Scenario *both*: mean change in top-1 with handedness evidence, random labels (calibrated, or scaled to the spread of the handedness log-odds) and random directions of HC-183.

errors unrelated to hand, so a known hand removes half of the competitors. The value of a predicted hand is set by Eq. (1): filtering needs about 87% accuracy and Bayesian fusion about 70% for a one-point gain, whereas writer-independent handedness from offline Bangla handwriting reaches 65.5–73.8% here and 77% balanced accuracy in [14]. Predicted handedness should therefore not filter galleries and should be fused only as a calibrated probability, which gives a rare minority class a small gain at little cost; known handedness is worth using in casework.

The random-label control shows why high attribute accuracies are easy to obtain and misleading: trained on the enrolled writers, a meaningless label reached up to 76.2%, more than real handedness for unseen writers. Soft-biometric studies should state whether test writers were seen in training, and a random-label run is a cheap check. Eq. (1) needs only two accuracies, so it can be applied to other attributes before any fusion experiment.

Identification used frozen and handcrafted features by design, as 42 writers are too few to train identification networks without a risk of overfitting. The main limitation is sample size, set by availability: left-handed writers are rare and left-hand use is culturally discouraged in many regions [22], so the balanced collection has 42 writers from one region and one phone. No other Bangla collection with handedness labels exists for replication, offline photographs cannot record pen pressure or stroke order, and per-writer gender is not in the metadata, so only one real attribute could be tested.

VII. Conclusion

On 42 Bangla writers, known handedness raised writer identification accuracy by 6.0 points, whereas handedness predicted for unseen writers (65.5% accurate) did not help and gallery filtering cost 14.4 points. A utility law predicted these results, and a random-label control showed that attribute accuracies obtained on enrolled writers mainly reflect writer identity. Soft biometrics should be fused with identification only when they are known or predicted far more accurately than offline handwriting currently allows. Future work will test Eq. (1) on further attributes, such as gender, and on

TABLE V

COMPARISON WITH PREVIOUS WORK. ✓: ADDRESSED; ×: NOT ADDRESSED ACCORDING TO THE CITED TITLE AND ABSTRACT; –: NOT APPLICABLE OR NOT STATED. ATTR. TO ID: ATTRIBUTE USED TO CHANGE WRITER IDENTIFICATION; KNOWN/PRED.: KNOWN AND PREDICTED ATTRIBUTE COMPARED; BREAK-EVEN: ATTRIBUTE ACCURACY AT WHICH FUSION HELPS; RANDOM LABELS: CONTROL FOR WRITER IDENTITY IN ATTRIBUTE ACCURACY; PREV.: ATTRIBUTE PREVALENCE IN THE GALLERY; ACQ.: ACQUISITION-CONTROLLED DATA. G: GENDER, H: HANDEDNESS, SB: SEVERAL SOFT BIOMETRICS.

Work	Script	Writers	Attr.	Writer ID	Attr. to ID	Known/ pred.	Break-even	Random labels	Prev.	Acq.	Key result
[1]	–	–	SB	✓	✓	–	–	–	–	–	–
[3]	–	–	G	×	×	×	×	×	×	–	–
[5]	–	–	G	×	×	×	×	×	×	–	competition, 194 teams
[4]	–	–	G	×	×	×	×	×	×	–	–
[6]	–	–	SB	×	×	×	×	×	×	–	–
[7]	Arabic, English	121	H	×	×	×	×	×	×	–	F-measure about 70%
[8]	English, Arabic	–	G, H	×	×	×	×	×	×	–	–
[9]	–	–	G, H	×	×	×	×	×	×	–	above 99%
[11]	–	–	–	✓	×	–	–	–	–	–	–
[17]	Bangla	100	–	✓	×	–	–	–	–	–	94.54% verification
[14]	Bangla	42	H	✓	×	×	×	×	×	✓	77.4% writer bal. acc.
This work	Bangla	42	H	✓	✓	✓	✓	✓	✓	✓	known +6.0, predicted −0.6 points; filter needs 87%

larger collections in other scripts, and will try to raise writer-independent handedness accuracy towards the break-even point, for example with more text per questioned document or with pen-pressure and on-line stroke cues.

DATA AND CODE AVAILABILITY

The BanglaHand dataset is publicly available under a CC BY 4.0 licence [10]. The analysis code and saved results will be released upon acceptance.

ETHICS STATEMENT

Informed consent was obtained voluntarily from all participants prior to data collection. No personal identifying information was collected, and all handwriting samples are pseudonymous.

REFERENCES

[1] A. K. Jain, S. C. Dass, and K. Nandakumar, "Soft biometric traits for personal recognition systems," in *Biometric Authentication (ICBA 2004)*, ser. Lecture Notes in Computer Science, vol. 3072. Springer, 2004, pp. 731–738.

[2] A. Dantcheva, P. Elia, and A. Ross, "What else does your biometric data reveal? A survey on soft biometrics," *IEEE Trans. Inf. Forensics Secur.*, vol. 11, no. 3, pp. 441–467, 2016.

[3] M. Liwicki, A. Schlapbach, and H. Bunke, "Automatic gender detection using on-line and off-line information," *Pattern Anal. Appl.*, vol. 14, no. 1, pp. 87–92, 2011.

[4] I. Siddiqi, C. Djeddi, A. Raza, and L. Souici-Meslati, "Automatic analysis of handwriting for gender classification," *Pattern Anal. Appl.*, vol. 18, no. 4, pp. 887–899, 2015.

[5] A. Hassaïne, S. Al-Maadeed, J. Aljaam, and A. Jaoua, "ICDAR 2013 competition on gender prediction from handwriting," in *Proceedings of the 12th International Conference on Document Analysis and Recognition (ICDAR)*. IEEE, 2013, pp. 1417–1421.

[6] N. Bouadjenek, H. Nemmour, and Y. Chibani, "Robust soft-biometrics prediction from off-line handwriting analysis," *Appl. Soft Comput.*, vol. 46, pp. 980–990, 2016.

[7] S. Al-Maadeed, F. Ferjani, S. Elloumi, and A. Jaoua, "A novel approach for handedness detection from off-line handwriting using fuzzy conceptual reduction," *EURASIP J. Image Video Process.*, vol. 2016, p. 1, 2016.

[8] Á. Morera, Á. Sánchez, J. F. Vélez, and A. B. Moreno, "Gender and handedness prediction from offline handwriting using convolutional neural networks," *Complexity*, vol. 2018, p. 3891624, 2018.

[9] M. Rahmanian and M. A. Shayegan, "Handwriting-based gender and handedness classification using convolutional neural networks," *Multimed. Tools Appl.*, vol. 80, pp. 35 341–35 364, 2021.

[10] T. H. Prappo, S. Hassan, S. Ali, S. N. Dhar, and M. S. Rahman, "BanglaHand," Mendeley Data, V1, 2026. [Online]. Available: https://doi.org/10.17632/n4sjstdzfy.1

[11] M. Bulacu and L. Schomaker, "Text-independent writer identification and verification using textural and allographic features," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 29, no. 4, pp. 701–717, 2007.

[12] L. Schomaker, K. Franke, and M. Bulacu, "Using codebooks of fragmented connected-component contours in forensic and historic writer identification," *Pattern Recognition Letters*, vol. 28, no. 6, pp. 719–727, 2007.

[13] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, "Leakage in data mining: Formulation, detection, and avoidance," *ACM Transactions on Knowledge Discovery from Data*, vol. 6, no. 4, pp. 15:1–15:21, 2012.

[14] Anonymous, "Writer-independent handedness identification from offline handwriting," Under review; details omitted for double-blind review, 2026.

[15] M. F. Mridha, A. Q. Ohi, M. A. Ali, M. I. Emon, and M. M. Kabir, "BanglaWriting: A multi-purpose offline Bangla handwriting dataset," *Data Brief*, vol. 34, p. 106633, 2021.

[16] N. Majid and E. H. Barney Smith, "Introducing the Boise State Bangla Handwriting Dataset and an efficient offline recognizer of isolated Bangla characters," in *2018 16th International Conference on Frontiers in Handwriting Recognition (ICFHR)*. IEEE, 2018, pp. 380–385.

[17] J. Paul, K. Dutta, A. Sarkar, K. Roy, and N. Das, "Writer verification using feature selection based on genetic algorithm: A case study on handwritten Bangla dataset," *ETRI Journal*, vol. 46, no. 4, pp. 648–659, 2024.

[18] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 770–778.

[19] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. 36th Int. Conf. Mach. Learn. (ICML)*, ser. Proc. Mach. Learn. Res., vol. 97, 2019, pp. 6105–6114.

[20] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16x16 words: Transformers for image recognition at scale," in *Int. Conf. Learn. Represent. (ICLR)*, 2021.

[21] J. C. Platt, "Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods," in *Advances in Large Margin Classifiers*, A. J. Smola, P. Bartlett, B. Schölkopf, and D. Schuurmans, Eds. Cambridge, MA: MIT Press, 1999, pp. 61–74.

[22] M. Papadatou-Pastou, E. Ntolka, J. Schmitz, M. Martin, M. R. Munafò, S. Ocklenburg, and S. Paracchini, "Human handedness: A meta-analysis," *Psychological Bulletin*, vol. 146, no. 6, pp. 481–524, 2020.